# 1.46 2D Vectors

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## Vector Basics

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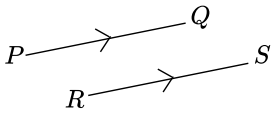
A Whereas a **coordinate** represents a **position** in space, a **vector** represents a **displacement** in space.

A vector has 2 properties:
- Direction
- Magnitude (i.e. length)

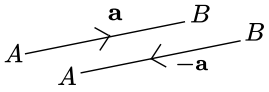
If $P$ and $Q$ are points then $\vec{PQ}$ is the vector between them.




B If two vectors $\vec{PQ}$ and $\vec{RS}$ have the same magnitude and direction, **they're the same vector** and are **parallel**.




C $\vec{AB} = -\vec{BA}$ and the two vectors are parallel, equal in magnitude but in **opposite directions**.




D Triangle Law for vector addition:

$$\vec{AB} + \vec{BC} = \vec{AC}$$




The vector of multiple vectors is known as the **resultant vector.**
(you will encounter this term in Mechanics)

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## Just for your interest…

4

Have you ever wondered what happens if you 'multiply' two vectors or two sets?

In KS2/3 you probably only experienced variables holding numerical values. You since saw that variables can represent other mathematical types, such as sets or vectors:

$$x = 3$$
$$\mathbf{a} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$
$$T = \{2, 5, 9, 10\}$$

'Multiplying' **two sets** (known again as the cross product) finds each possible combination of members, one from each:

$$\{a, b, c\} \times \{d, e\} = \{\{a, d\}, \{a, e\}, \{b, d\}, \{b, e\}, \{c, d\}, \{c, e\}\}$$

It has the nice property that $n(A \times B) = n(A) \times n(B)$, where $n(A)$ gives the size of the set $A$. Also, $A \times B = B \times A$.

But when we do we need to define **explicitly** what operators like '$+$' and '$\times$' mean.

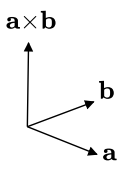
$$1 + 3 = 4 \qquad \begin{pmatrix} 3 \\ -2 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$$

$$\begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = ? \qquad \{a, b, c\} \times \{d, e\} = ?$$

Often these operators are defined to give it properties that are consistent with its usage elsewhere, e.g. 'commutativity': $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$ for vector addition just as $2 + 4 = 4 + 2$ for numbers.

In FM, you will see that 'multiplying' two 3D vectors (known as the **cross product**) gives you a vector **perpendicular to the two**.
Vector multiplication is not commutative, so $\mathbf{a} \times \mathbf{b} \neq \mathbf{b} \times \mathbf{a}$ (however it is 'distributive', so $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{c})$)
The Casio Classwiz can calculate this in Vector mode!

$$\begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix}$$



## 1.46 2D Vectors

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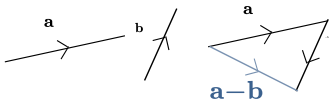
# Vector Basics

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E

Vector **subtraction** is defined using vector addition and negation:

$$\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b})$$




F

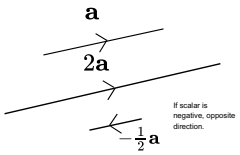
The zero vector $\mathbf{0}$ (a bold 0), represents no movement.

$$\vec{PQ} + \vec{QP} = \mathbf{0}$$

In 2D: $\mathbf{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

G

A **scalar** is a normal number, which can be used to 'scale' a vector.
- The **direction** will be the **same**.
- But the **magnitude** will be **different** (unless the scalar is 1).




H

Any vector parallel to the vector $\mathbf{a}$ can be written as $\lambda\mathbf{a}$, where $\lambda$ is a scalar.

The implication is that if we can write one vector **as a multiple of** another, then we can show they are parallel.

"Show $2\mathbf{a} + 4\mathbf{b}$ and $3\mathbf{a} + 6\mathbf{b}$ are parallel".
$3\mathbf{a} + 6\mathbf{b} = \frac{3}{2}(\mathbf{a} + 2\mathbf{b}) \therefore$ parallel

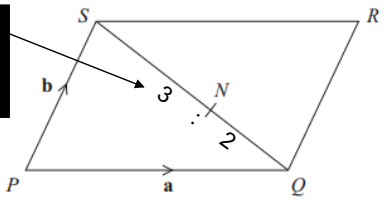
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# Example

6

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**Tip**: This ratio wasn't in the original diagram. I like to add the ratio as a visual aid.




$PQRS$ is a parallelogram.

$N$ is the point on $SQ$ such that $SN : NQ = 3 : 2$.

$\overrightarrow{PQ} = \mathbf{a}$ $\overrightarrow{PS} = \mathbf{b}$

(a) Write down, in terms of $\mathbf{a}$ and $\mathbf{b}$, an expression for $\overrightarrow{SQ}$.

(b) Express $\overrightarrow{NR}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

a

For (b), there's two possible paths to get from N to R: via S or via Q. But which is best?

b

$\vec{QR}$ is also $b$ because it is exactly the same movement as $\vec{PS}$.

## 1.46 2D Vectors

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# Test Your Understanding

7

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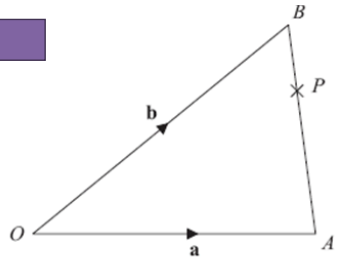



Diagram **NOT** accurately drawn

$OAB$ is a triangle.

$\overrightarrow{OA} = \mathbf{a}$
$\overrightarrow{OB} = \mathbf{b}$

(a) Find $\overrightarrow{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

.....
(1)

$P$ is the point on $AB$ such that $AP : PB = 3 : 1$

(b) Find $\overrightarrow{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.
Give your answer in its simplest form.

.....

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# Exercise 11A - Year 1 Pages 234-235

8

**6** $ABCD$ is a trapezium with $AB$ parallel to $DC$ and $DC = 3AB$. $M$ divides $DC$ such that $DM : MC = 2 : 1$. $\overrightarrow{AB} = \mathbf{a}$ and $\overrightarrow{BC} = \mathbf{b}$. Find, in terms of $\mathbf{a}$ and $\mathbf{b}$: **a** $\overrightarrow{AM}$\quad**b** $\overrightarrow{BD}$\quad**c** $\overrightarrow{MB}$\quad**d** $\overrightarrow{DA}$.

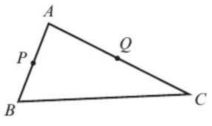
**9** In triangle $ABC$, $\overrightarrow{AB} = \mathbf{a}$ and $\overrightarrow{AC} = \mathbf{b}$.

$P$ is the midpoint of $AB$ and $Q$ is the midpoint of $AC$.

**a** Write in terms of $\mathbf{a}$ and $\mathbf{b}$:

**i** $\overrightarrow{BC}$ \quad**ii** $\overrightarrow{AP}$ \quad**iii** $\overrightarrow{AQ}$ \quad**iv** $\overrightarrow{PQ}$

**b** Show that $PQ$ is parallel to $BC$.



## 1.46 2D Vectors

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### Exercise 11A - Year 1 Pages 234-235

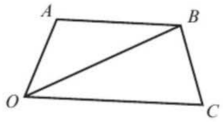
9

**10** $OABC$ is a quadrilateral. $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OC} = 3\mathbf{b}$ and $\overrightarrow{OB} = \mathbf{a} + 2\mathbf{b}$.

**a** Find, in terms of $\mathbf{a}$ and $\mathbf{b}$:

i $\overrightarrow{AB}$ \quad ii $\overrightarrow{CB}$

**b** Show that $AB$ is parallel to $OC$.

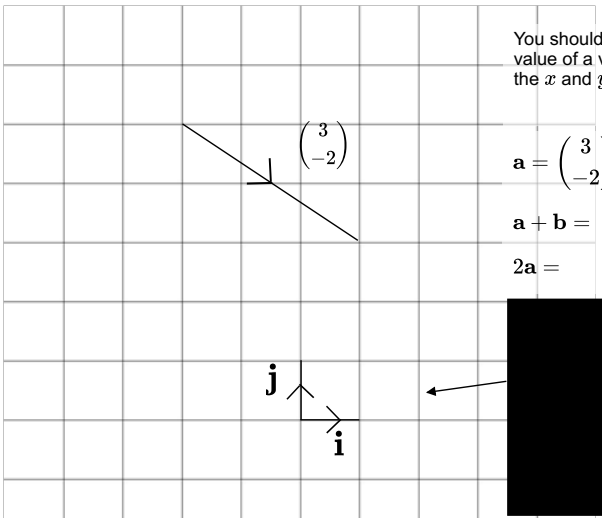



**11** The vectors $2\mathbf{a} + k\mathbf{b}$ and $5\mathbf{a} + 3\mathbf{b}$ are parallel. Find the value of $k$.

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### Representing Vectors

10




You should already be familiar that the value of a vector is the **displacement** in the $x$ and $y$ direction (if in 2D).

$\mathbf{a} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$

$\mathbf{a} + \mathbf{b} =$

$2\mathbf{a} =$

[redacted]

**Notes**: This allows us to write any vector algebraically without using vector notation. **Any point in 2D space,** as a vector from the origin, can be obtained using a linear combination of $\mathbf{i}$ and $\mathbf{j}$, e.g. if $P(5, -1)$, $\overrightarrow{OP} = 5\mathbf{i} - \mathbf{j}$. For this reason, $\mathbf{i}$ and $\mathbf{j}$ are known as **basis vectors** of 2D coordinate space. In fact, any two non-parallel/non-zero vectors can be used as basis vectors, e.g. if $\mathbf{a} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$, it's possible to get any vector $\begin{pmatrix} x \\ y \end{pmatrix}$ using a linear combination of these, i.e. we can always find scalars $p$ and $q$ such that $\begin{pmatrix} x \\ y \end{pmatrix} = p\begin{pmatrix} 5 \\ 2 \end{pmatrix} + q\begin{pmatrix} 3 \\ 0 \end{pmatrix}$.

## 1.46 2D Vectors

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# Examples

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If $\mathbf{a} = 3\mathbf{i}, \quad \mathbf{b} = \mathbf{i} + \mathbf{j}, \quad \mathbf{c} = \mathbf{i} - 2\mathbf{j}$ then:

1) Write $\mathbf{a}$ in vector form.

2) Find $\mathbf{b}+\mathbf{2c}$ in $\mathbf{i}$,$\mathbf{j}$ form.

1

2

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# Exercise 11B - Year 1 Pages 237-238

12

**4** Given that $\mathbf{a} = 2\mathbf{i} + 5\mathbf{j}$ and $\mathbf{b} = 3\mathbf{i} - \mathbf{j}$, find:
**a** $\lambda$ if $\mathbf{a} + \lambda\mathbf{b}$ is parallel to the vector $\mathbf{i}$.
**b** $\mu$ if $\mu\mathbf{a} + \mathbf{b}$ is parallel to the vector $\mathbf{j}$.

**5** Given that $\mathbf{c} = 3\mathbf{i} + 4\mathbf{j}$ and $\mathbf{d} = \mathbf{i} - 2\mathbf{j}$, find:
**a** $\lambda$ if $\mathbf{c} + \lambda\mathbf{d}$ is parallel to $\mathbf{i} + \mathbf{j}$.
**b** $\mu$ if $\mu\mathbf{c} + \mathbf{d}$ is parallel to $\mathbf{i} + 3\mathbf{j}$.
**c** $s$ if $\mathbf{c} - s\mathbf{d}$ is parallel to $2\mathbf{i} + \mathbf{j}$.
**d** $t$ if $\mathbf{d} - t\mathbf{c}$ is parallel to $-2\mathbf{i} + 3\mathbf{j}$.

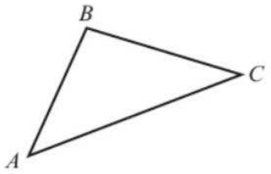
## 1.46 2D Vectors

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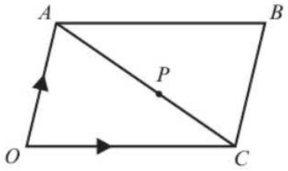
### Exercise 11B - Year 1 Pages 237-238

13

**6** In triangle $ABC$, $\overrightarrow{AB} = 4\mathbf{i} + 3\mathbf{j}$ and $\overrightarrow{AC} = 5\mathbf{i} + 2\mathbf{j}$. Find $\overrightarrow{BC}$.




**7** $OABC$ is a parallelogram. $P$ divides $AC$ in the ratio $3 : 2$. $\overrightarrow{OA} = 2\mathbf{i} + 4\mathbf{j}$, $\overrightarrow{OC} = 7\mathbf{i}$. Find in $\mathbf{i}, \mathbf{j}$ format and column vector format: **a** $\overrightarrow{AC}$\quad**b** $\overrightarrow{OP}$\quad**c** $\overrightarrow{AP}$.




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### Exercise 11B - Year 1 Pages 237-238

14

**8**\quad $\mathbf{a} = \begin{pmatrix} j \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 10 \\ k \end{pmatrix}$, $\mathbf{c} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$. Given that $\mathbf{b} - 2\mathbf{a} = \mathbf{c}$, find the values of $j$ and $k$.

**9** $\mathbf{a} = \begin{pmatrix} p \\ -q \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} q \\ p \end{pmatrix}$, $\mathbf{c} = \begin{pmatrix} 7 \\ 4 \end{pmatrix}$

Given that $\mathbf{a} + 2\mathbf{b} = \mathbf{c}$, find the values of $p$ and $q$.

**10** The resultant of the vectors $\mathbf{a} = 3\mathbf{i} - 2\mathbf{j}$ and $\mathbf{b} = p\mathbf{i} - 2p\mathbf{j}$ is parallel to the vector $\mathbf{c} = 2\mathbf{i} - 3\mathbf{j}$. Find:
**a** the value of $p$ (4 m
**b** the resultant of vectors $\mathbf{a}$ and $\mathbf{b}$. (1

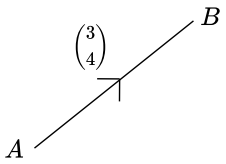
## 1.46 2D Vectors

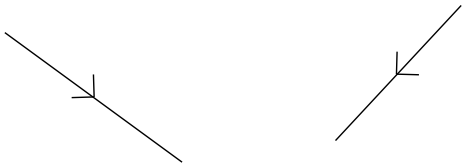
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# Magnitude of a Vector

15

The magnitude $|a|$ of a vector $a$ is its length.



$\mathbf{a} = \begin{pmatrix}4\\-1\end{pmatrix}$ $|\mathbf{a}| =$

$\mathbf{b} = \begin{pmatrix}2\\0\end{pmatrix}$ $|\mathbf{b}| =$

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# Unit Vectors

16

A unit vector is a vector whose magnitude is 1

There's certain operations on vectors that require the vectors to be 'unit' vectors. We just scale the vector so that its magnitude is now 1.

$$\mathbf{a} = \begin{pmatrix}3\\4\end{pmatrix}$$

$$|\mathbf{a}| =$$

**Test Your Understanding:** Convert the following vectors to unit vectors.

$\mathbf{a} = \begin{pmatrix}12\\-5\end{pmatrix}$ $\mathbf{b} = \begin{pmatrix}1\\1\end{pmatrix}$

$|\mathbf{a}| =$ $|\mathbf{b}| =$

$\therefore \hat{\mathbf{a}} =$ $\therefore \hat{\mathbf{b}} =$

## 1.46 2D Vectors

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### Exercise 11C - Year 1 Pages 240-242

17

**8** Given that $|2\mathbf{i} - k\mathbf{j}| = 2\sqrt{10}$, find the exact value of $k$.

**9** Vector $\mathbf{a} = p\mathbf{i} + q\mathbf{j}$ has magnitude 10 and makes an angle $\theta$ with the positive $x$-axis where $\sin\theta = \frac{3}{5}$. Find the possible values of $p$ and $q$.

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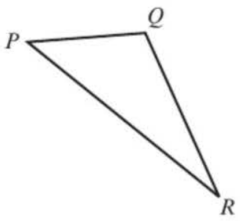
### Exercise 11C - Year 1 Pages 240-242

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**11** In triangle $PQR$, $\overrightarrow{PQ} = 4\mathbf{i} + \mathbf{j}$, $\overrightarrow{PR} = 6\mathbf{i} - 8\mathbf{j}$.

**a** Find the size of $\angle QPR$, in degrees, to one decimal place. \hfill (5 marks)

**b** Find the area of triangle $PQR$. \hfill (2 marks)



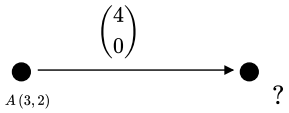
## 1.46 2D Vectors

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# Position Vectors

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Suppose we started at a point $(3, 2)$ and translated by the vector $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$:

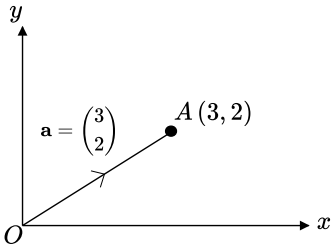



You might think we can do something like:

$$(3, 2) + \begin{pmatrix} 4 \\ 0 \end{pmatrix} = (7, 2)$$

But only vectors can be added to other vectors.
**If we treated the point $(3,2)$ as a vector**, then this solves the problem:

$$\begin{pmatrix} 3 \\ 2 \end{pmatrix} + \begin{pmatrix} 4 \\ 0 \end{pmatrix} = \begin{pmatrix} 7 \\ 2 \end{pmatrix}$$




A vector used to represent a position is unsurprisingly known as a **position vector**.
A position can be thought of as a translation from the origin, as per above. It enables us to use positions in all sorts of vector (and matrix!) calculations.

The position vector of a point $A$ is the vector $\vec{OA}$, where $O$ is the origin. $\vec{OA}$ is usually written as $\mathbf{a}$.

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# Example

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The points $A$ and $B$ have coordinates $(3, 4)$ and $(11, 2)$ respectively.
Find, in terms of $i$ and $j$:
a) The position vector of $A$
b) The position vector of $B$
c) The vector $\vec{AB}$

## 1.46 2D Vectors

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### Further Example

21

$\vec{OA} = 5i - 2j$ and $\vec{AB} = 3i + 4j$. Find:
a) The position vector of $B$.
b) The exact value of $\left|\vec{OB}\right|$ in simplified surd form.

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### Exercise 11D - Year 1 Pages 243-244

22

**4** $OABCDE$ is a regular hexagon. The points $A$ and $B$ have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, where $O$ is the origin. Find, in terms of $\mathbf{a}$ and $\mathbf{b}$, the position vectors of **a** $C$\quad**b** $D$\quad**c** $E$.

**5** *The position vectors of 3 vertices of a parallelogram* are $\begin{pmatrix}4\\2\end{pmatrix}$, $\begin{pmatrix}3\\5\end{pmatrix}$ and $\begin{pmatrix}8\\6\end{pmatrix}$.

Find the possible position vectors of the fourth vertex.

## 1.46 2D Vectors

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# Exercise 11D - Year 1 Pages 243-244

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**6** Given that the point $A$ has position vector $4\mathbf{i} - 5\mathbf{j}$ and the point $B$ has position vector $6\mathbf{i} + 3\mathbf{j}$,

**a** find the vector $\overrightarrow{AB}$. (2 ma

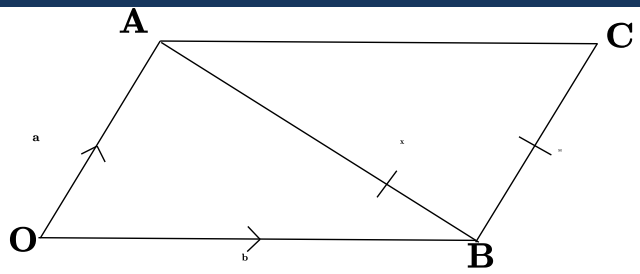
**b** find $|\overrightarrow{AB}|$ giving your answer as a simplified surd. (2 ma

**7** The point $A$ lies on the circle with equation $x^2 + y^2 = 9$. Given that $\overrightarrow{OA} = 2k\mathbf{i} + k\mathbf{j}$, find the exact value of $k$.

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# Solving Geometric Problems

24




$X$ is a point on $AB$ such that $AX : XB = 3 : 1$. $M$ is the midpoint of $BC$.
Show that $\vec{XM}$ is parallel to $\vec{OC}$.

For any proof question always find the vectors involved first, in this case $\vec{XM}$ and $\vec{OC}$.

The key is to factor out a scalar such that we see the same vector.

The magic words here are "*is a multiple of*".

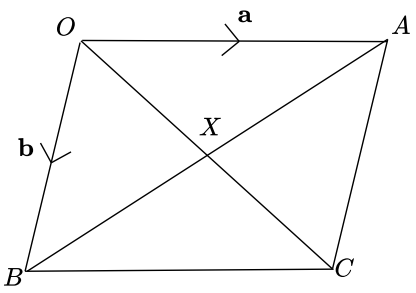
## 1.46 2D Vectors

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### Introducing Scalars and Comparing Coefficients

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Remember when we had **identities** like: $ax^2 + 3x \equiv 2x^2 + bx$
we could **compare coefficients**, so that $a = 2$ and $3 = b$.
**We can do the same with (non-parallel) vectors!**




$OACB$ is a parallelogram, where $\vec{OA} = a$ and $\vec{OB} = b$. The diagonals $OC$ and $AB$ intersect at a point $X$. Prove that the diagonals bisect each other.
*(Hint: Perhaps find $\vec{OX}$ in two different ways?)*

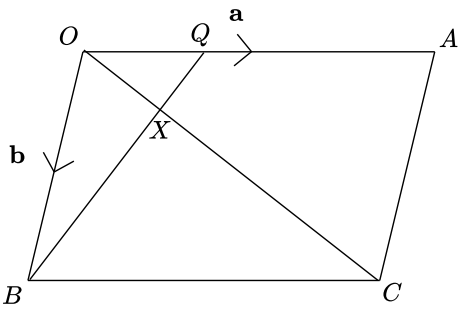
We don't know what fraction of the way across $\overrightarrow{BA}$ the point $X$ is, so let $\lambda$ be the fraction of the way across. We're hoping that $\lambda = \frac{1}{2}$.

We need to use a different scalar constant, this time $\mu$.
It is common to use the letters $\lambda$ and $\mu$ for scalars.

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### Test Your Understanding

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In the above diagram, $\vec{OA} = \mathbf{a}$, $\vec{OB} = \mathbf{b}$ and $\vec{OQ} = \frac{1}{3}\mathbf{a}$. We wish to find the ratio $OX : XC$.
a) If $\vec{OX} = \lambda\vec{OC}$, find an expression for $\vec{OX}$ in terms of $\mathbf{a}$,$\mathbf{b}$ and $\lambda$.
b) If $\vec{BX} = \mu\vec{BQ}$, find an expression for $\vec{OX}$ in terms of $\mathbf{a}$,$\mathbf{b}$ and $\mu$.
c) By comparing coefficients or otherwise, determine the value of $\lambda$, and hence the ratio $OX : XC$.

Expand and collect $\mathbf{a}$ terms and collect $\mathbf{b}$ terms, so that we can compare coefficients later.

## 1.46 2D Vectors

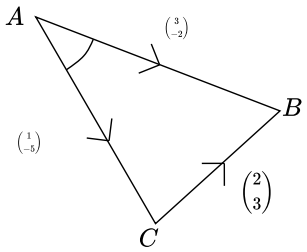
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### Area of a Triangle

27

$\vec{AB} = 3\mathbf{i} - 2\mathbf{j}$ and $\vec{AC} = \mathbf{i} - 5\mathbf{j}$. Determine $\angle BAC$.

**Strategy**: Find 3 lengths of triangle then use cosine rule to find angle.

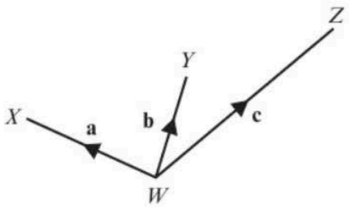



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### Exercise 11E - Year 1 Pages 246-247

28

**1** In the diagram, $\overrightarrow{WX} = \mathbf{a}$, $\overrightarrow{WY} = \mathbf{b}$ and $\overrightarrow{WZ} = \mathbf{c}$. It is given that $\overrightarrow{XY} = \overrightarrow{YZ}$.
Prove that $\mathbf{a} + \mathbf{c} = 2\mathbf{b}$.



## 1.46 2D Vectors

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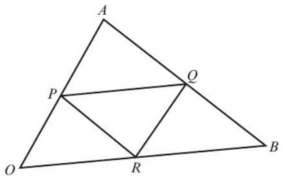
### Exercise 11E - Year 1 Pages 246-247

29

**2** $OAB$ is a triangle. $P$, $Q$ and $R$ are the midpoints of $OA$, $AB$ and $OB$ respectively. $OP$ and $OR$ are equal to $\mathbf{p}$ and $\mathbf{r}$ respectively.

**a** Find **i** $\overrightarrow{OB}$\quad**ii** $\overrightarrow{PQ}$.

**b** Hence prove that triangle $PAQ$ is similar to triangle $OAB$.




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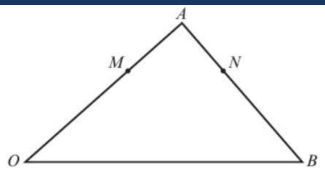
### Exercise 11E - Year 1 Pages 246-247

30

**3** $OAB$ is a triangle. $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. The point $M$ divides $OA$ in the ratio $2 : 1$. $MN$ is parallel to $OB$.

**a** Express the vector $\overrightarrow{ON}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

**b** Show that $AN : NB = 1 : 2$.



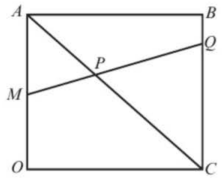
## 1.46 2D Vectors

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### Exercise 11E - Year 1 Pages 246-247

31

**4** $OABC$ is a square. $M$ is the midpoint of $OA$, and $Q$ divides $BC$ in the ratio $1:3$. $AC$ and $MQ$ meet at $P$.




**a** If $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OC} = \mathbf{c}$, express $\overrightarrow{OP}$ in terms of $\mathbf{a}$ and $\mathbf{c}$.
**b** Show that $P$ divides $AC$ in the ratio $2:3$.

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### Exercise 11E - Year 1 Pages 246-247

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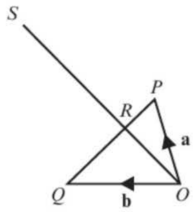
**6** $OPQ$ is a triangle.

$2\overrightarrow{PR} = \overrightarrow{RQ}$ and $3\overrightarrow{OR} = \overrightarrow{OS}$

$\overrightarrow{OP} = \mathbf{a}$ and $\overrightarrow{OQ} = \mathbf{b}$.

**a** Show that $\overrightarrow{OS} = 2\mathbf{a} + \mathbf{b}$.

**b** Point $T$ is added to the diagram such that $\overrightarrow{OT} = -\mathbf{b}$.




Prove that points $T$, $P$ and $S$ lie on a straight line.

t

## 1.46 2D Vectors

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# EXTENSION

33

1 *[STEP 2010 Q7]*

Relative to a fixed origin $O$, the points $A$ and $B$ have position vectors $\mathbf{a}$ and $\mathbf{b}$, respectively. (The points $O$, $A$ and $B$ are not collinear.) The point $C$ has position vector $\mathbf{c}$ given by

$$\mathbf{c} = \alpha\mathbf{a} + \beta\mathbf{b},$$

where $\alpha$ and $\beta$ are positive constants with $\alpha + \beta < 1$. The lines $OA$ and $BC$ meet at the point $P$ with position vector $\mathbf{p}$ and the lines $OB$ and $AC$ meet at the point $Q$ with position vector $\mathbf{q}$. Show that

$$\mathbf{p} = \frac{\alpha a}{1-\beta}$$

and write down $q$ in terms of $\alpha$, $\beta$ and $b$.

Show further that the point $R$ with position vector $r$ given by

$$\mathbf{r} = \frac{\alpha\mathbf{a}+\beta\mathbf{b}}{\alpha+\beta},$$

lies on the lines $OC$ and $AB$.

The lines $OB$ and $PR$ intersect at the point $S$. Prove that $\frac{OQ}{BQ} = \frac{OS}{BS}$.

Click here for the solution:
http://www.mathshelper.co.uk/STEP%202010%20Solutions.pdf
(go to Q7)

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# Modelling

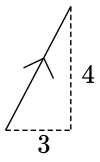
34

In Mechanics, you will see certain things can be represented as a simple number (without direction), or as a vector (with direction):

Remember a 'scalar' just means a normal number (in the context of vectors). It can be obtained using the **magnitude** of the vector.

| **Vector Quantity** | **Equivalent Scalar Quantity** |
|---|---|
| Velocity<br>e.g. $\begin{pmatrix}3\\4\end{pmatrix} km/h$ | Speed<br>$= 5km/h$ |
| Displacement<br>e.g. $\begin{pmatrix}-5\\12\end{pmatrix} km$ | |

This means the position vector of the object changes by $\begin{pmatrix}3\\4\end{pmatrix}$ each hour.




5

…which is equivalent to moving 5km each hour.

## 1.46 2D Vectors

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### Example

35

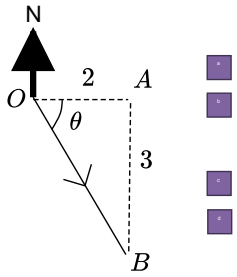
[Textbook] A girl walks 2 km due east from a fixed point $O$ to $A$, and then 3 km due south from $A$ to $B$. Find

a) the total distance travelled

b) the position vector of $B$ relative to $O$

c) $\left|\vec{OB}\right|$

d) The bearing of $B$ from $O$.




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### Further Example

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[Textbook] In an orienteering exercise, a cadet leaves the starting point $O$ and walks 15 km on a bearing of $120^\circ$ to reach $A$, the first checkpoint. From $A$ he walks 9 km on a bearing of $240^\circ$ to the second checkpoint, at $B$. From $B$ he returns directly to $O$.

Find:

a) the position vector of $A$ relative to $O$

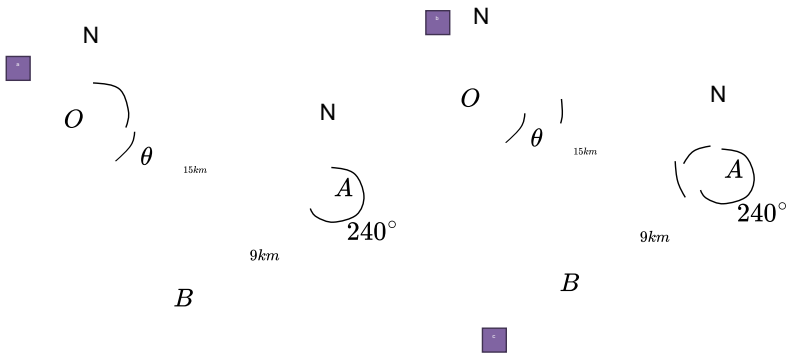
b) $\left|\vec{OB}\right|$

c) the bearing of $B$ from $O$

d) the position vector of $B$ relative $O$.

I have no specific advice to offer except:

1. Draw a BIG diagram.
2. Remember bearings are measured clockwise from North.
3. Don't forget units (on vectors!)



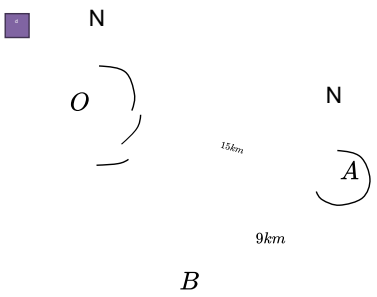
## 1.46 2D Vectors

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# Further Example

37

[Textbook] In an orienteering exercise, a cadet leaves the starting point $O$ and walks 15 km on a bearing of $120^\circ$ to reach $A$, the first checkpoint. From $A$ he walks 9 km on a bearing of $240^\circ$ to the second checkpoint, at $B$. From $B$ he returns directly to $O$.
Find:
a) the position vector of $A$ relative to $O$
b) $\left|\vec{OB}\right|$
c) the bearing of $B$ from $O$
d) the position vector of $B$ relative $O$.




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# Exercise 11F - Year 1 Pages 250-251

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**5** A particle $P$ of mass $m = 0.3$ kg moves under the action of a single constant force $\mathbf{F}$ newtons. The acceleration of $P$ is $\mathbf{a} = (5\mathbf{i} + 7\mathbf{j})\ \mathrm{m\ s^{-2}}$.
**a** Find the angle between the acceleration and $\mathbf{i}$. (2 ma
Force, mass and acceleration are related by the formula $\mathbf{F} = m\mathbf{a}$.
**b** Find the magnitude of $\mathbf{F}$. (3 ma

**6** Two forces, $\mathbf{F}_1$ and $\mathbf{F}_2$, are given by the vectors $\mathbf{F}_1 = (3\mathbf{i} - 4\mathbf{j})$ N and $\mathbf{F}_2 = (p\mathbf{i} + q\mathbf{j})$ N. The resultant force, $\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2$ acts in a direction which is parallel to the vector $(2\mathbf{i} - \mathbf{j})$.

**a** Find the angle between $\mathbf{R}$ and the vector $\mathbf{i}$.

**b** Show that $p + 2q = 5$.

**c** Given that $p = 1$, find the magnitude of $\mathbf{R}$.

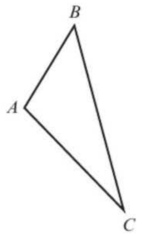
## 1.46 2D Vectors

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### Exercise 11F - Year 1 Pages 250-251

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**7** The diagram shows a sketch of a field in the shape of a triangle $ABC$. Given $\overrightarrow{AB} = 30\mathbf{i} + 40\mathbf{j}$ metres and $\overrightarrow{AC} = 40\mathbf{i} - 60\mathbf{j}$ metres,




**a** find $\overrightarrow{BC}$ (2 ma
**b** find the size of $\angle BAC$, in degrees, to one decimal place (4 ma
**c** find the area of the field in square metres. (3 ma

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### Exercise 11F - Year 1 Pages 250-251

40

**8** A boat has a position vector of $(2\mathbf{i} + \mathbf{j})$ km and a buoy has a position vector of $(6\mathbf{i} - 4\mathbf{j})$ km, relative to a fixed origin $O$.
**a** Find the distance of the boat from the buoy.
**b** Find the bearing of the boat from the buoy.
The boat travels with constant velocity $(8\mathbf{i} - 10\mathbf{j})$ km/h.
**c** Verify that the boat is travelling directly towards the buoy.
**d** Find the speed of the boat.
**e** Work out how long it will take the boat to reach the buoy.
**Problem-solving** Draw a sketch showing t
initial positions of the bo
the buoy and the origin.